

Kyle Young

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EDUCATION

University of California, Los Angeles

2024 – 2028

B.S. in Mechanical Engineering — GPA: 3.83;

Los Angeles, CA

- Deans Honor List, Samueli Foundation Centennial Scholar; Richard L. Gay Trust Scholar, Samueli Research Scholar
- Coursework: Statics, Dynamics, Strength of Materials, Thermodynamics, Fluid Mechanics, Mechanisms and Mechanical Systems, C++ Programming

PROFESSIONAL EXPERIENCE

Northrop Grumman

June 2025 – Aug 2025

Engineering Intern, Module Assembly and Test Group

Manhattan Beach, CA

- Designed a semi-automated cleanroom support robot to replace manual manufacturing workflows, improving training efficiency by 80%, reducing machine downtime by 7×, and eliminating FOD risk.
- Led qualification of a replacement test material for flight hardware by planning and executing 10+ validation tests, analyzing results, and producing aerospace-standard documentation.
- Improved module assembly and test operations by identifying process bottlenecks, supporting production hardware handling, and documenting repeatable manufacturing procedures.

Atomic Machines

June 2023 – Aug 2023

Engineering Intern, AMP Engineering Team

Berkeley, CA

- Designed and fabricated production hardware for a MEMS manufacturing platform using CAD, metal laser cutting, manual machining, and rapid prototyping.
- Ran reliability testing on a platform-wide pick-and-place subsystem, identifying failure modes and improving measured reliability by 30%.
- Developed Python calibration scripts to improve measurement accuracy, reduce manual test setup time, and streamline engineering validation workflows.

LEADERSHIP AND TECHNICAL PROJECTS

Bruin Formula Racing (FSAE EV)

Sep 2024 – Present

Suspension Lead

Los Angeles, CA

- Lead 5 responsible engineers through the design and development of the car's suspension system, coordinating subsystem goals, design reviews, manufacturing timelines, and subteam integration
- Previously, owned the shock packaging subsystem, designing damper, bellcrank, and mount geometry around packaging, clearance, motion ratio, and serviceability requirements.
- Built kinematic studies to evaluate motion ratios, keeping to vehicle dynamics targets within 5% deviation.
- Reduced bellcrank weight by 12.5% by converting MATLAB load cases into FEA-driven design iterations, topology optimization, manufacturing drawings, and machined components.

Complex Fluids and Interfacial Physics Laboratory

June 2026 – Present

Undergraduate Researcher, PI: Prof. Pirouz Kavehpour

Los Angeles, CA

- Analyze cerebrospinal fluid flow data to support a noninvasive method for evaluating whether Chiari malformation surgery improved patient flow dynamics.
- Develop Python scripts to extract, clean, and compare physiological flow features from patient datasets in collaboration with the David Geffen School of Medicine at UCLA.

SF Unity Robotics — FIRST Team 9038

Feb 2022 – Jun 2024

Founder and Mentor

San Francisco, CA

- Founded and scaled a 50+ student independent FRC team, building the organization from initial recruitment through two awards at the FIRST World Championship.
- Led mechanical design and fabrication for two 125 lb competition robots, using CAD, prototyping, laser cutting, and manual machining to iterate from concept to competition-ready systems.
- Raised \$55K through 20+ successful grants from sponsors including NASA and Apple, funding robot development, team operations, and competition travel.

SKILLS

CAD/CAE: Onshape, SolidWorks, Fusion 360, PTC Creo, FEA, topology optimization, GD&T

Manufacturing: CAD/CAM, DFM, 3D printing, laser cutting, manual mill, manual lathe, prototyping, test design

Programming: MATLAB, Python, C++